Chapter 6
Physics of MOS Transistors

Sun-Ting Lin
林尚亭

Electronics and Photonics Post-undergraduate Second Degree Program
National Yang Ming Chiao Tung University

Chapter 6 Physics of MOS Transistors

- 6.1 Structure of MOSFET
- 6.2 Operation of MOSFET
- 6.3 MOS Device Models
- 6.4 PMOS Transistor
- 6.5 CMOS Technology
- 6.6 Comparison of Bipolar and CMOS Devices

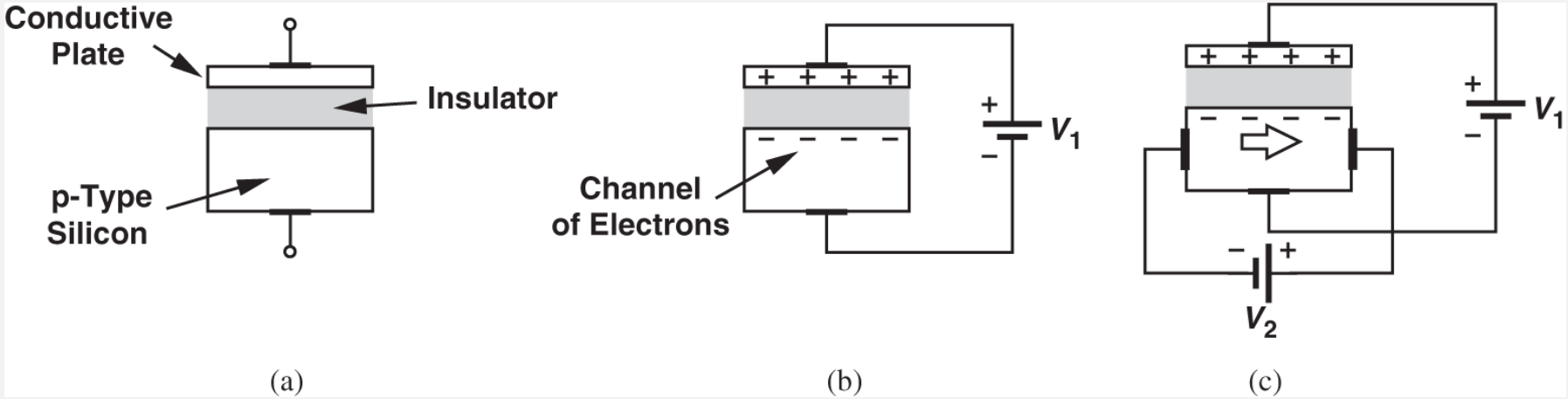
Outline

- The outline below illustrates the sequence of concepts covered in this chapter.

Operation of MOSFETs	MOS Device Models	PMOS Devices
• MOS Structure • Operation in Triode Region • Operation in Saturation • I/V Characteristics	• Large−Signal Model • Small−Signal Model	• Structure • Models

Metal-Oxide-Semiconductor (MOS) Capacitor

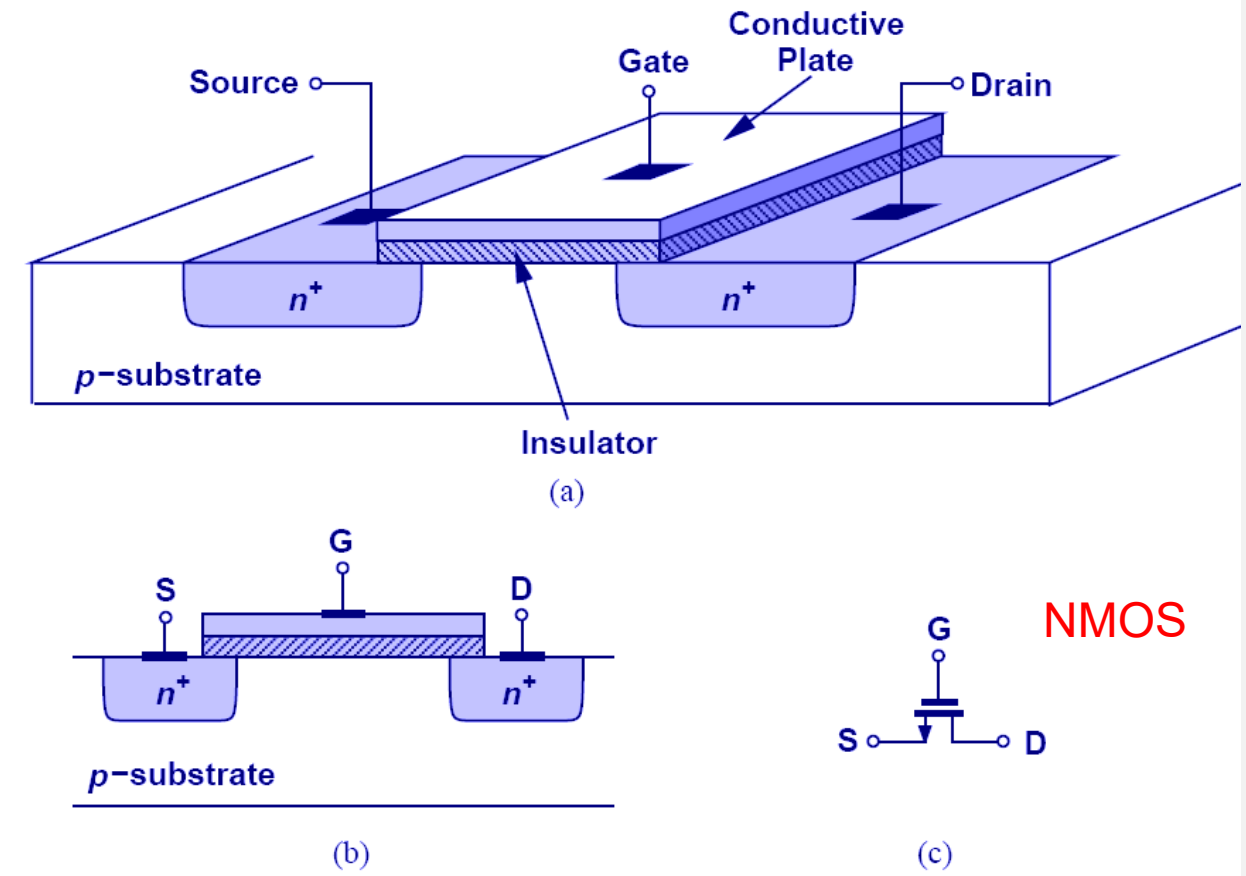
- The MOS structure can be thought of as a parallel-plate capacitor, with the top plate being the positive plate, oxide being the dielectric, and Si substrate being the negative plate.
- Equation Q = CV suggests that, to achieve a strong control of Q by V, the value of C must be maximized, for example, by reducing the thickness of the dielectric layer separating the two plates.



(a) Hypothetical semiconductor device, (b) operation as a capacitor, (c) current flow as a result of potential difference.

Structure and Symbol of MOSFET

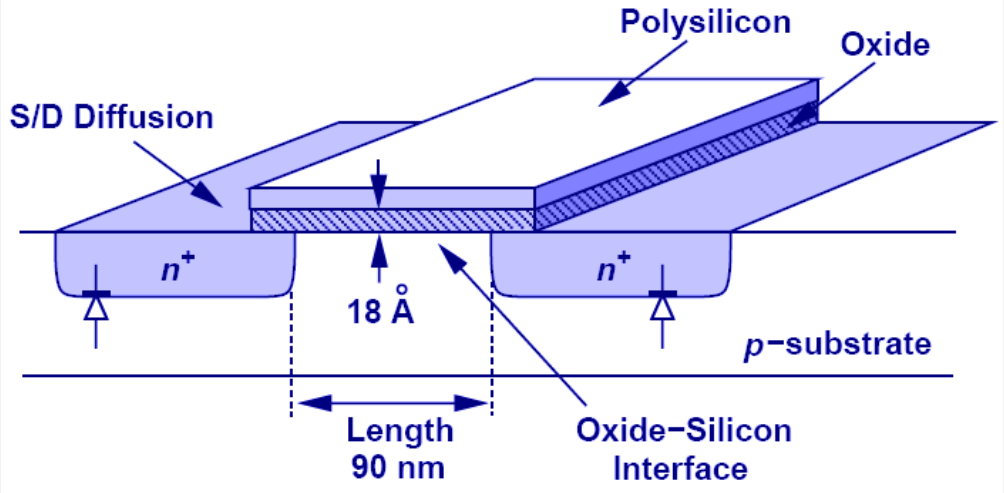
- This device is symmetric, so either of the n^+ regions can be source or drain.



(a) Structure of MOSFET, (b) side view, (c) circuit symbol.

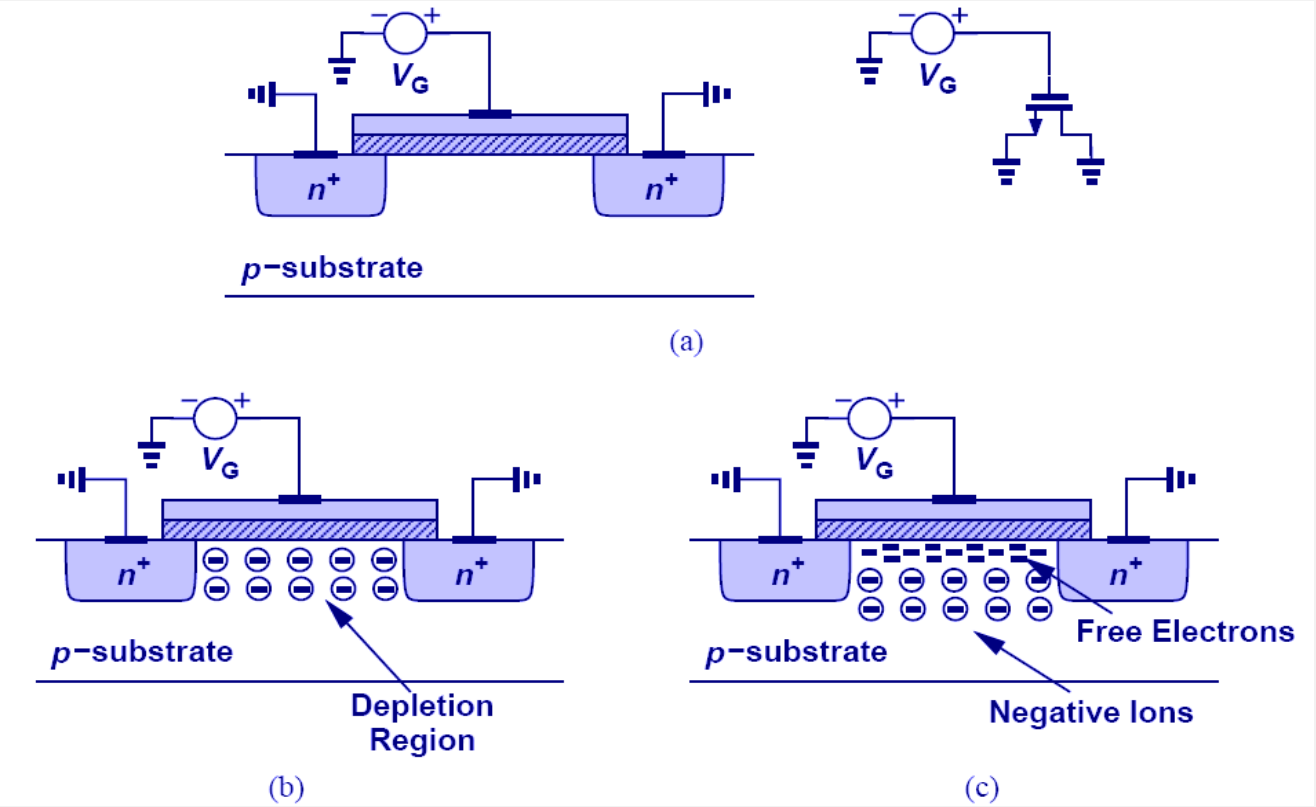
State of the Art MOSFET Structure

- The gate is formed by low-resistivity polysilicon (or simply “poly”), and the insulator by silicon dioxide (or simply “oxide”).
- The dielectric layer sandwiched between the gate and the substrate plays a critical role in the performance of transistors.



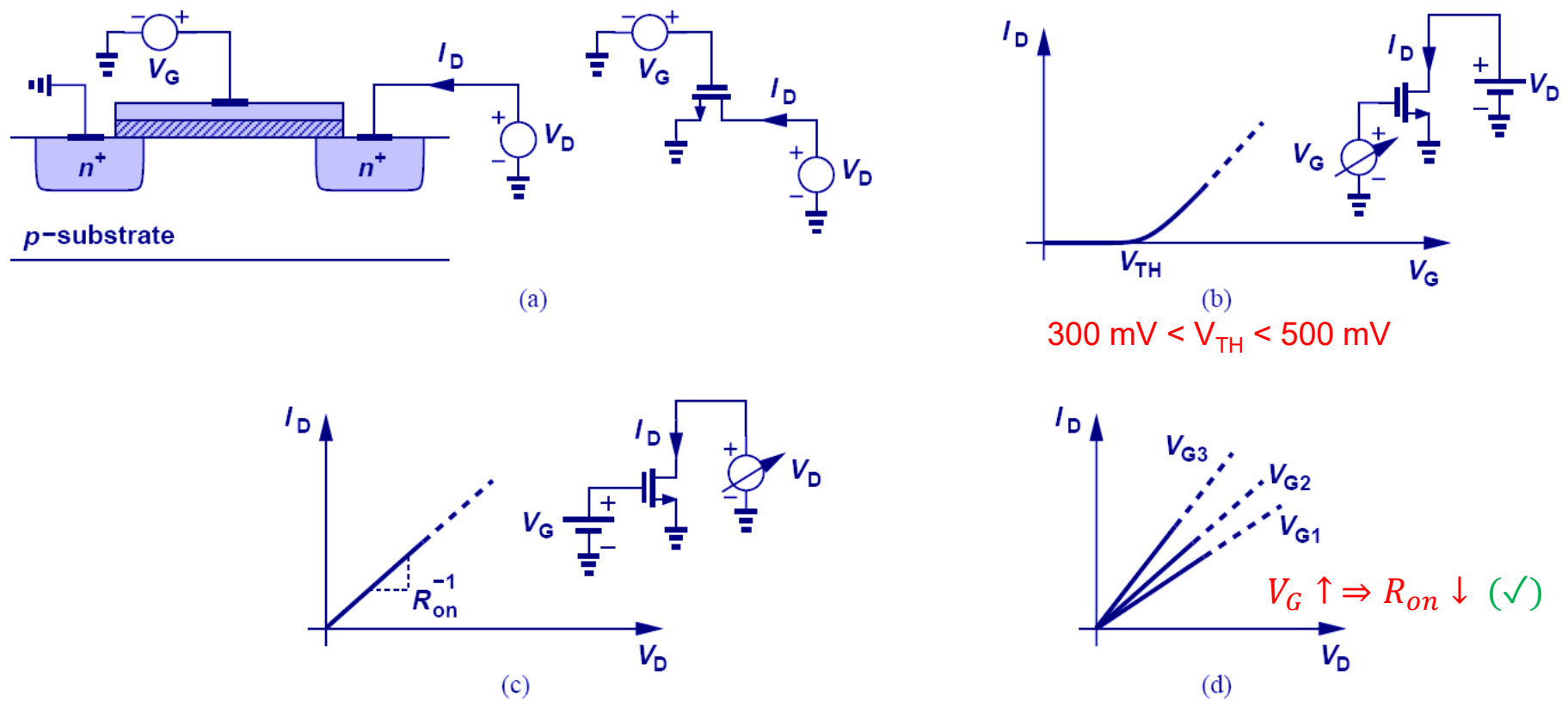
Formation of Channel

- First, the holes are repelled by the positive gate voltage, leaving behind negative ions and forming a depletion region. Next, electrons are attracted to the interface, creating a channel ("inversion layer").
- The transport mechanism in a MOSFET is the drift of charge.



MOSFET Characteristics: I_D-V_G and I_D-V_D

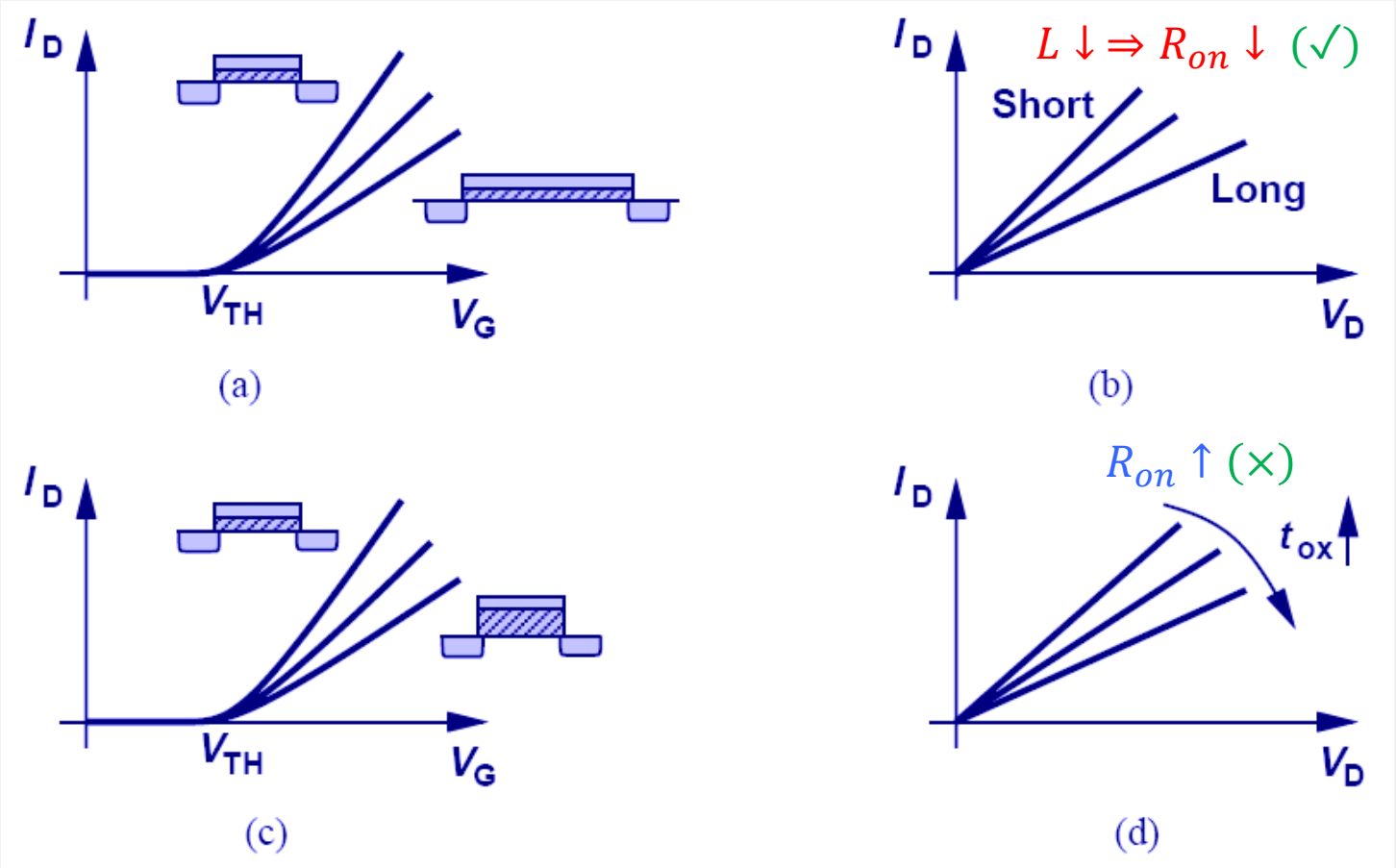
- The MOS characteristics are measured by varying V_G while keeping V_D constant, and varying V_D while keeping V_G constant.



(a) MOSFET with gate and drain voltages, (b) I_D-V_G characteristic, (c) I_D-V_D characteristic, (d) voltage-dependent resistor: I_D-V_D characteristics for various gate voltages.

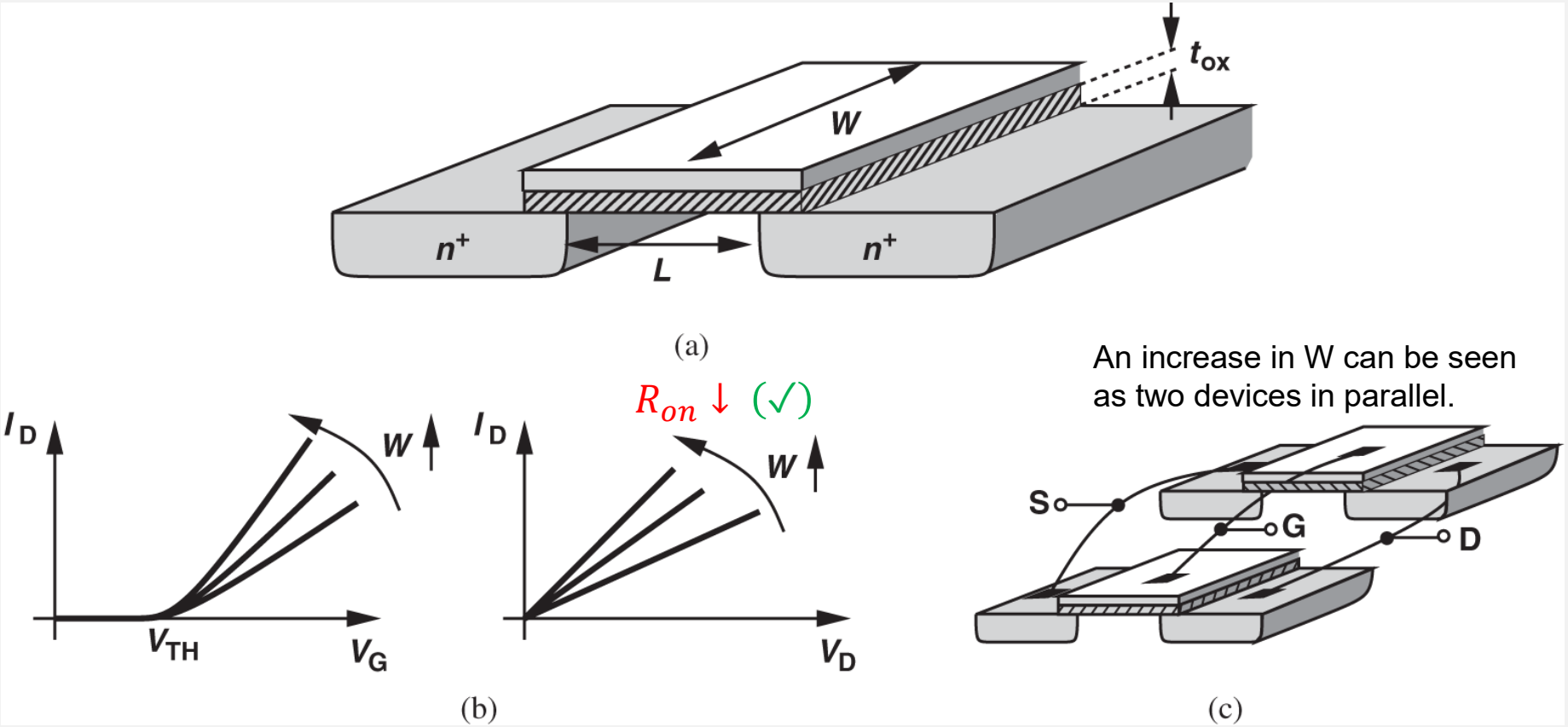
Gate Length and Oxide Thickness and Dependence

- $L \downarrow$ or $t_{ox} \downarrow \Rightarrow R_{on} \downarrow$ (✓): Small gate length (L) and oxide thickness (t_{ox}) yield low channel resistance, which will increase the drain current.



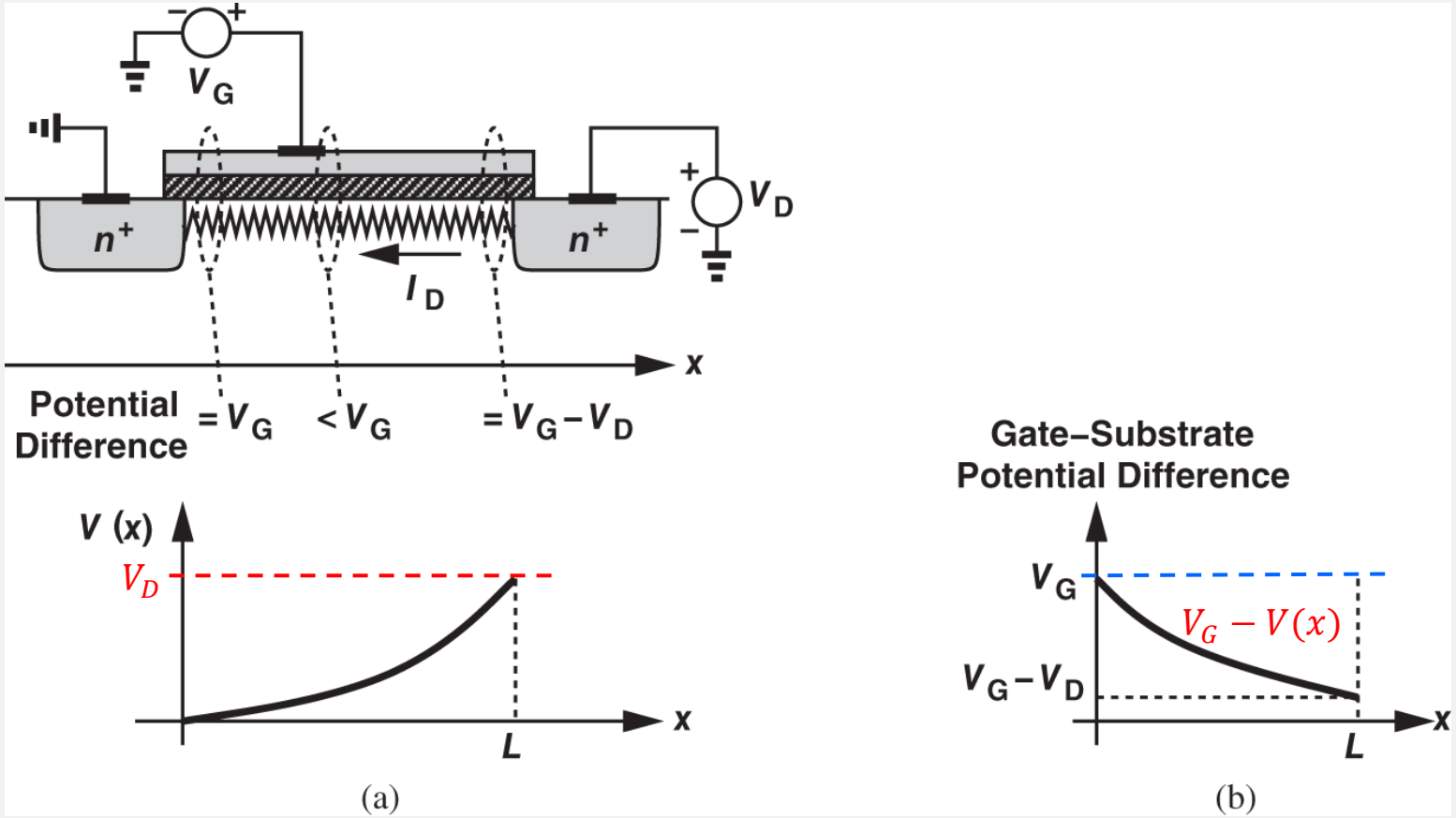
Effect of Gate Width

- $W \uparrow \Rightarrow R_{on} \downarrow (\checkmark)$, $C \uparrow \Rightarrow Speed \downarrow (\times)$: As the gate width (W) increases, the current increases due to a decrease in resistance. However, gate capacitance also increases thus, limiting the speed of the circuit.



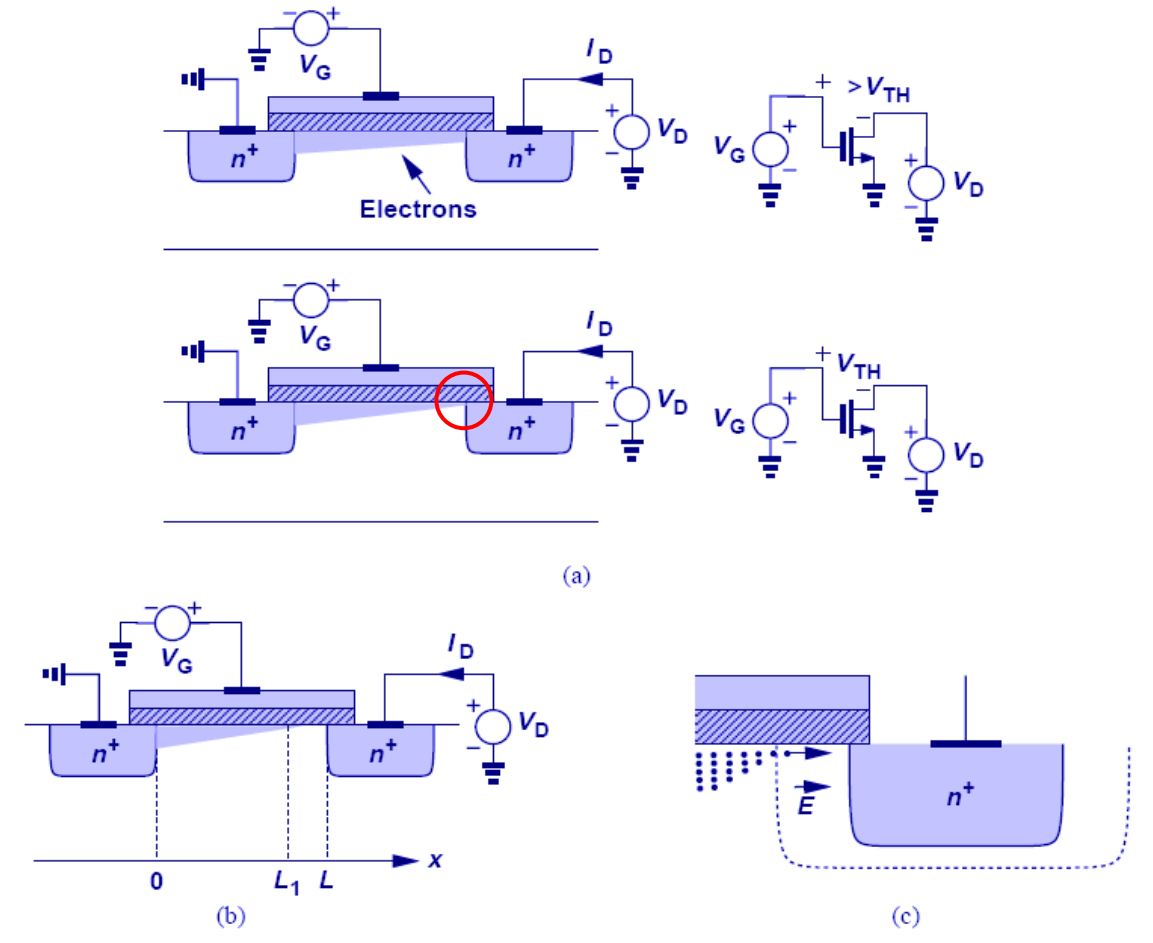
Channel Potential Variation

- Since there's a channel resistance between drain and source, and if drain is biased higher than the source, channel potential increases from source to drain, and the potential between gate and channel (the oxide-silicon interface) will decrease from source to drain.



Channel Pinch-Off

- As the potential difference between drain and gate becomes more positive, the inversion layer beneath the interface starts to pinch off around drain.



$V_{GD} > V_{TH}$: more electrons in the channel.

(a) $V_{GD} = V_{TH}$: the channel at drain totally pinches off (V_{GD} is not sufficient at x = L to attract electrons)

(b) When $(V_{GD} < V_{TH})$, $V_D \uparrow \Rightarrow L_1 \downarrow$: the channel length starts to decrease.

(c) When $(V_{GD} < V_{TH})$: The device still conducts. The electrons experience the high electric field and rapidly swept to the drain.

Channel Charge Density

- The channel charge (i.e., free electrons) density is equal to the gate capacitance times the gate voltage in excess of the threshold voltage.

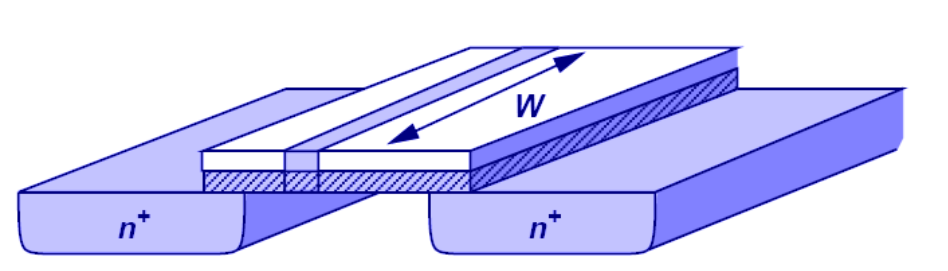
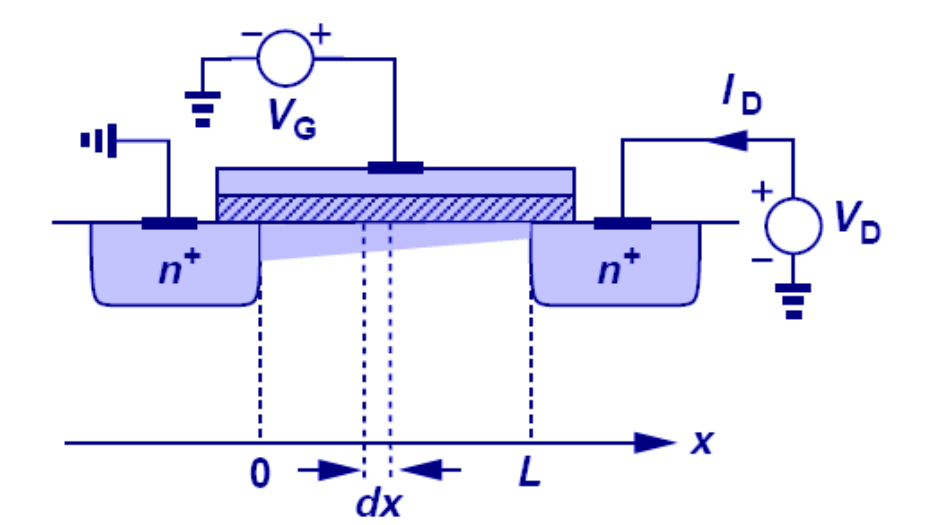


Illustration of capacitance per unit length.



Device illustration for calculation of drain current.

$Q = CV$

$C = WC_{ox}$, where C_{ox} is the gate capacitance per unit area.

$V = (V_{GS} - V_{TH})$

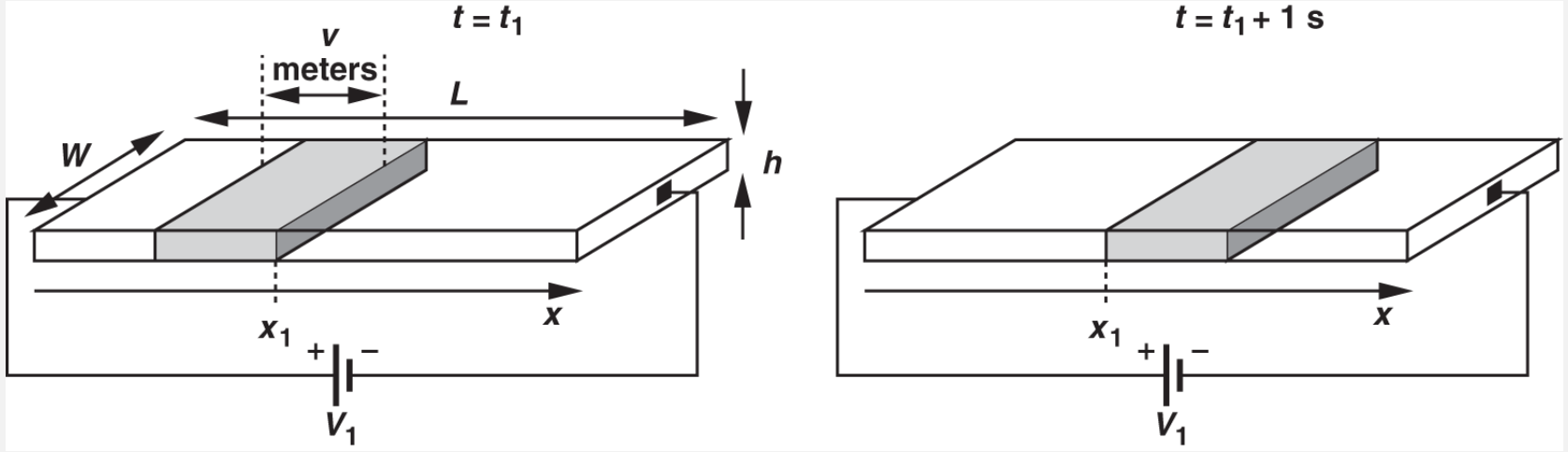
$Q = WC_{ox}(V_{GS} - V_{TH})$

Let x be a point along the channel from source to drain, and V(x) its potential; the expression above gives the charge density (per unit length).

$Q(x) = WC_{ox}[V_{GS} - V(x) - V_{TH}]$

Charge Density and Current

- The current that flows from source to drain (electrons) is related to the charge density in the channel by the charge velocity v (m/s).



$$I = Q \cdot v$$

Drain Current I_D

$$v = -\mu_n E = \mu_n \frac{dV}{dx}$$

$$I_D = Q \cdot v$$
$$= WC_{ox}[V_{GS} - V(x) - V_{TH}] \cdot \mu_n \frac{dV(x)}{dx}$$

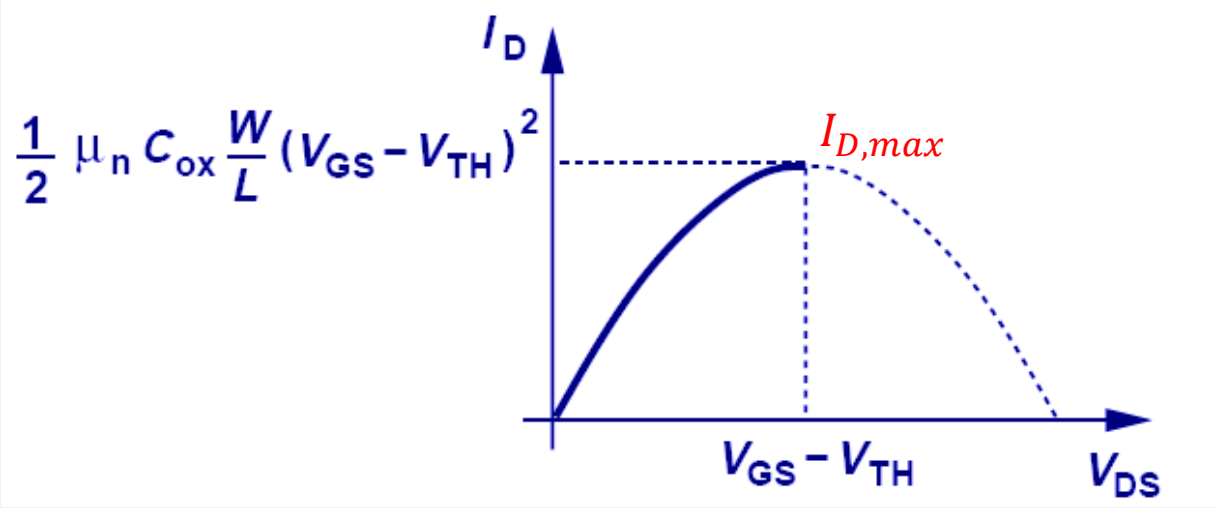
$$\vdots$$

$$I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}[2(V_{GS} - V_{TH})V_{DS} - V_{DS}^2]$$

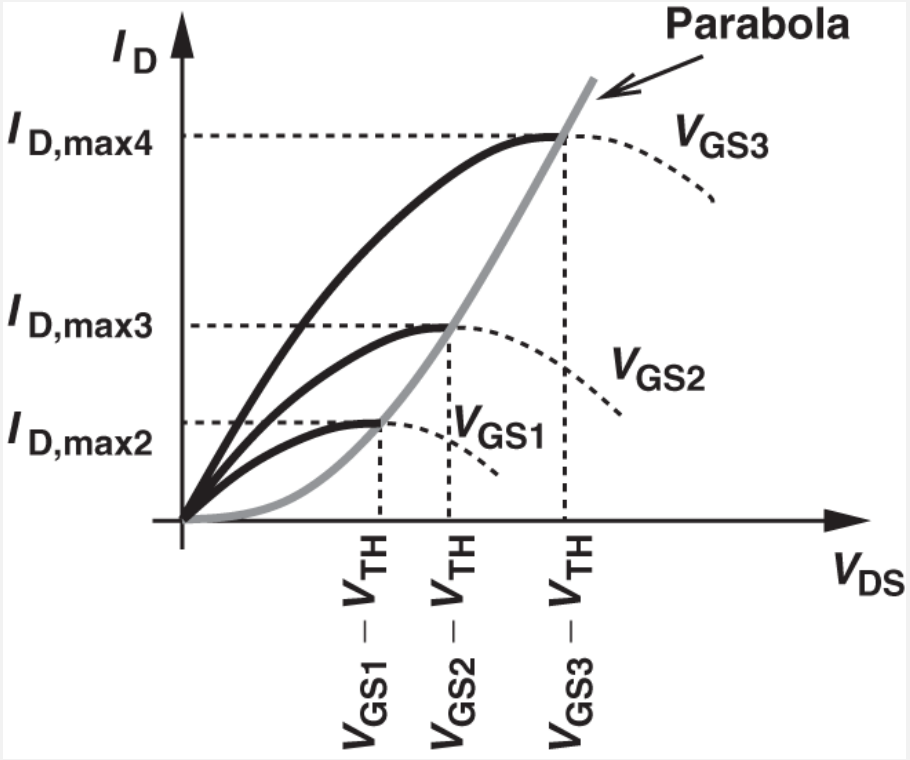
$$I_{D,max} = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS} - V_{TH})^2$$

Parabolic I_D-V_{DS} Relationship

- By keeping V_{GS} constant and varying V_{DS}, we obtain a parabolic relationship.
- The maximum current $(I_{D,max})$ occurs when V_{DS} equals to V_{GS}- V_{TH}.



I_D-V_{DS} for Different Values of V_{GS}

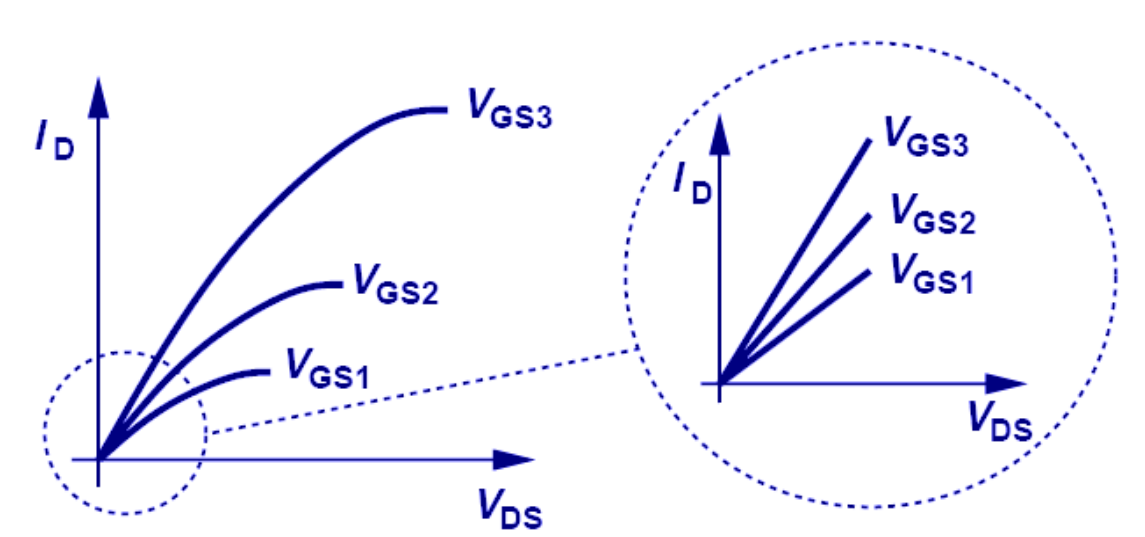


$I_{D,max} \propto (V_{GS} - V_{TH})^2$

$V_{GS} \uparrow \Rightarrow (V_{GS} - V_{TH}) \uparrow \Rightarrow I_{D,max}$

Linear Resistance

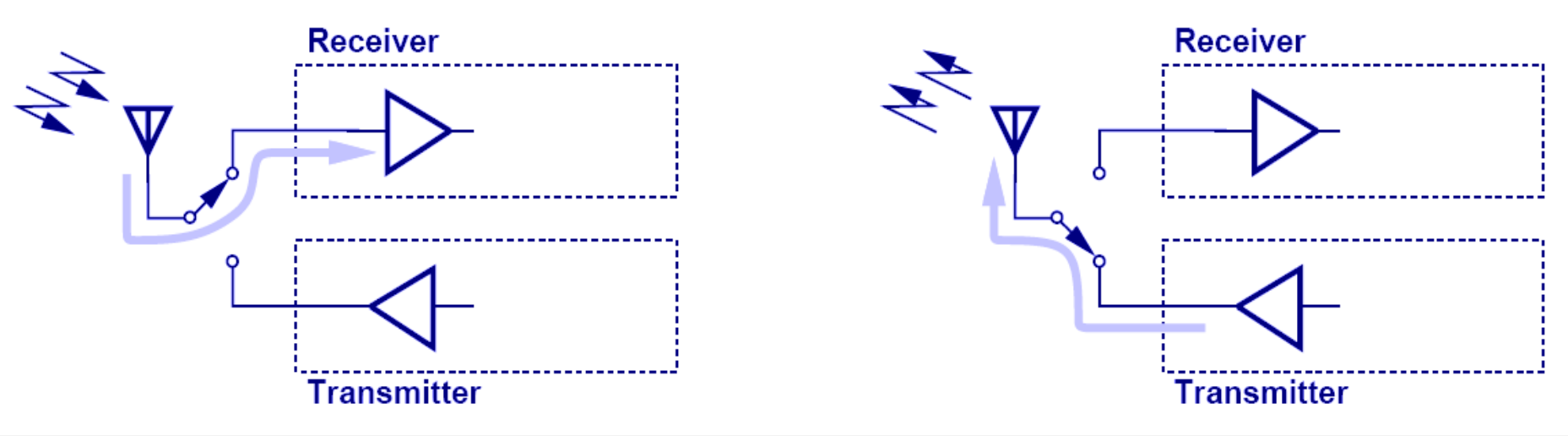
- At small V_{DS}, the transistor can be viewed as a resistor, with the resistance depending on the gate voltage.
- It finds application as an electronic switch.



$$R_{on} = \frac{1}{\mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})}$$

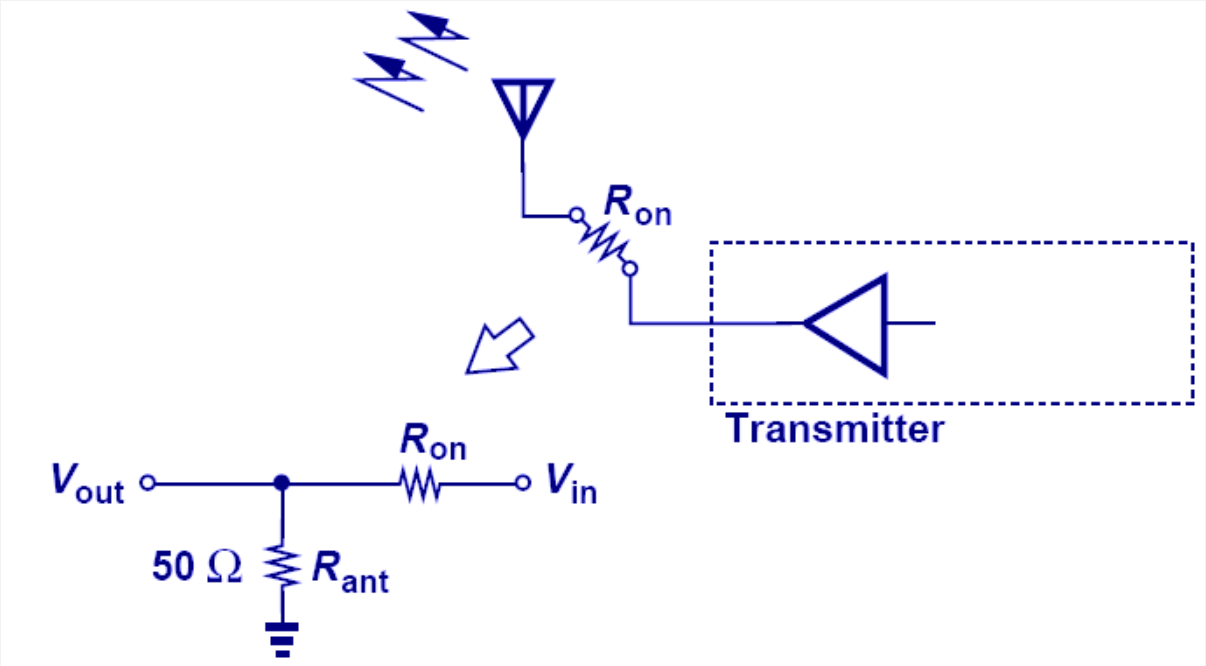
Application of Electronic Switches

- In a cordless telephone system in which a single antenna is used for both transmission and reception, a switch is used to connect either the receiver or transmitter to the antenna.

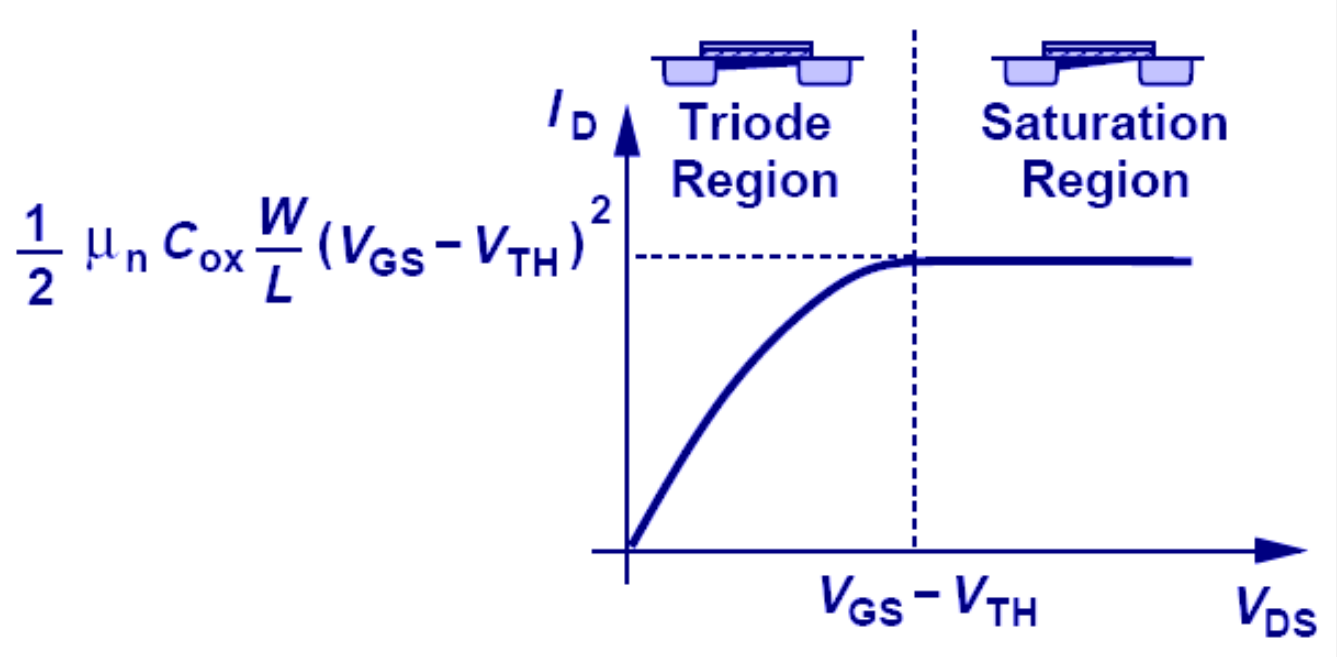


Effects of On-Resistance

- To minimize signal attenuation, R_{on} of the switch has to be as small as possible. This means larger W/L aspect ratio and greater V_{GS}.

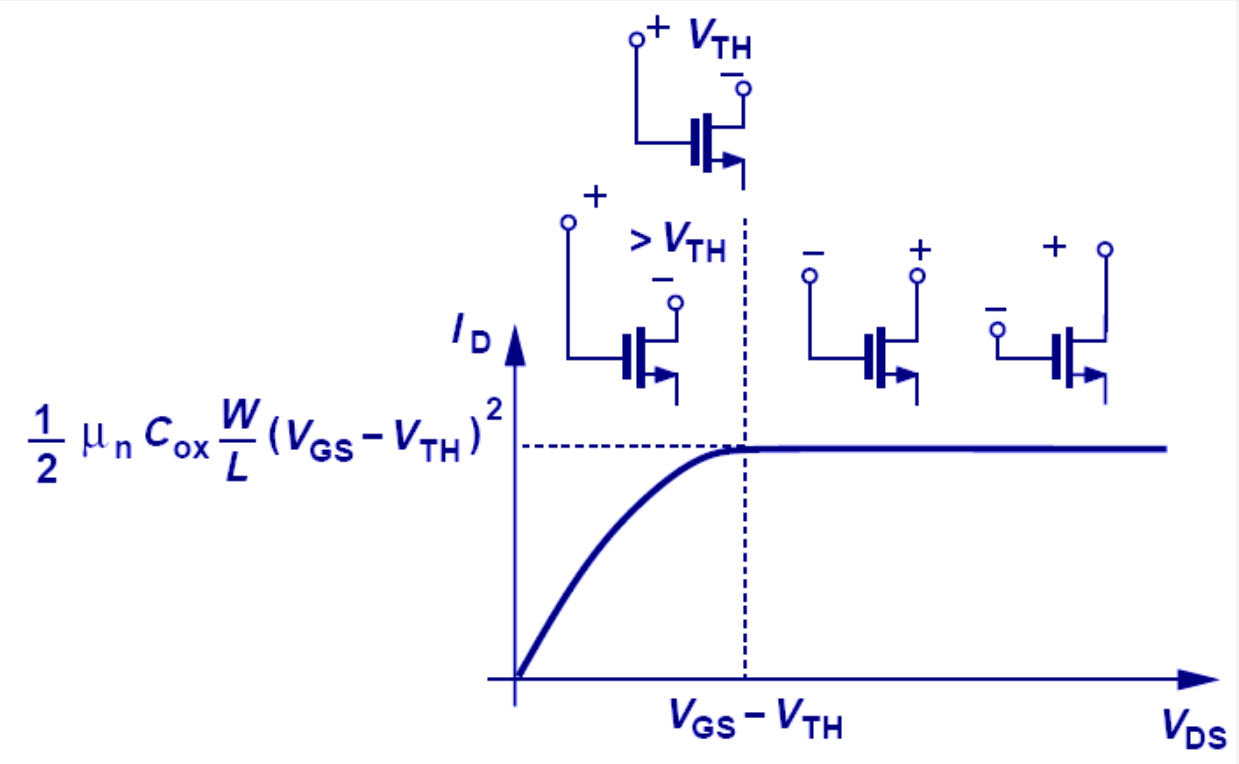


Different Regions of Operation



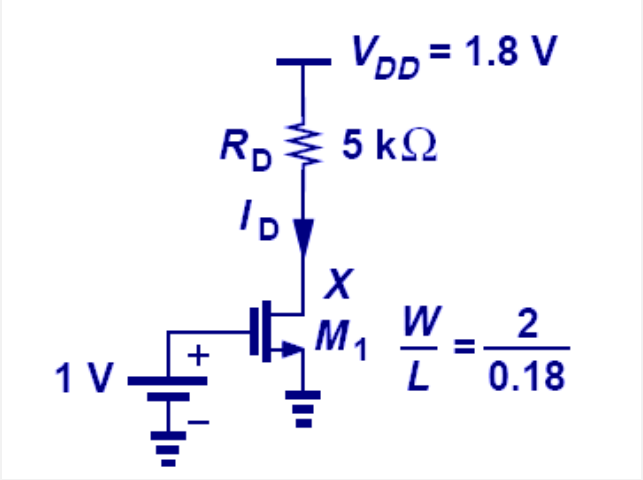
How to Determine ‘Region of Operation’

- $V_{GD} < V_{TH}$ (saturation region): When the potential difference between gate and drain becomes equal to or less than V_{TH}, the MOSFET enters saturation region.
- $V_{GD} > V_{TH}$ (triode region): When the potential difference between gate and drain is greater than V_{TH}, the MOSFET is in triode region.



Example 6-6

- Calculate the bias current. Assume $\mu_n C_{ox} = 100\,\mu\text{A/V}^2$ and V_{TH} = 0.4 V. If the gate voltage increases by 10 mV, what is the change in the drain voltage?



- **Solution**:

Assume M_1 is in saturation region for $V_{GS} = 1\ \text{V}$.

$$I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS} - V_{TH})^2$$
$$= \frac{1}{2}(100 \times 10^{-6})\frac{2}{0.18}(1 - 0.4)^2$$
$$= 200 \times 10^{-6}$$

$$V_{DS} = V_{DD} - I_D R_D$$
$$= 1.8 - (200 \times 10^{-6}) \times (5 \times 10^3)$$
$$= 0.8\ \text{V}$$

$$V_{GD} = V_{GS} - V_{DS} = 1 - 0.8 = 0.2$$

$(V_{GD} = 0.2) < (V_{TH} = 0.4)$, the M_1 is in the saturation region.

Assume M_1 is in saturation region for $V_{GS} = 1.01\ \text{V}$.

$$I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS} - V_{TH})^2$$
$$= \frac{1}{2}(100 \times 10^{-6})\frac{2}{0.18}(1.01 - 0.4)^2$$
$$= 207 \times 10^{-6}$$

$$V_{DS} = V_{DD} - I_D R_D$$
$$= 1.8 - (207 \times 10^{-6}) \times (5 \times 10^3)$$
$$= 0.765\ \text{V}$$

$$V_{GD} = V_{GS} - V_{DS} = 1 - 0.765 = 0.235$$

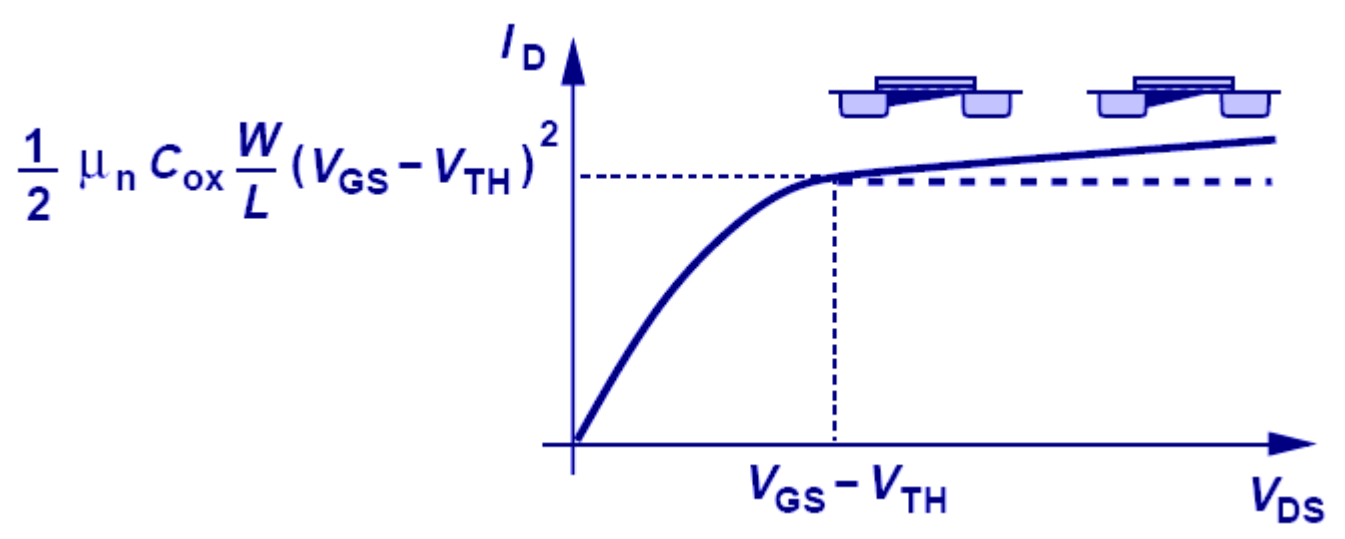
$(V_{GD} = 0.235) < (V_{TH} = 0.4)$, the M_1 is still in the saturation region.

$$\triangle V_{GS} = 10\ \text{mV} \Longrightarrow \triangle V_{DS} = 35\ \text{mV}$$

The circuit can *amplify* the input.

Channel-Length Modulation

- The original observation that the current is constant in the saturation region is not quite correct. The end point of the channel actually moves toward the source as V_D increases, increasing I_D. Therefore, the current in the saturation region is a weak function of the drain voltage.

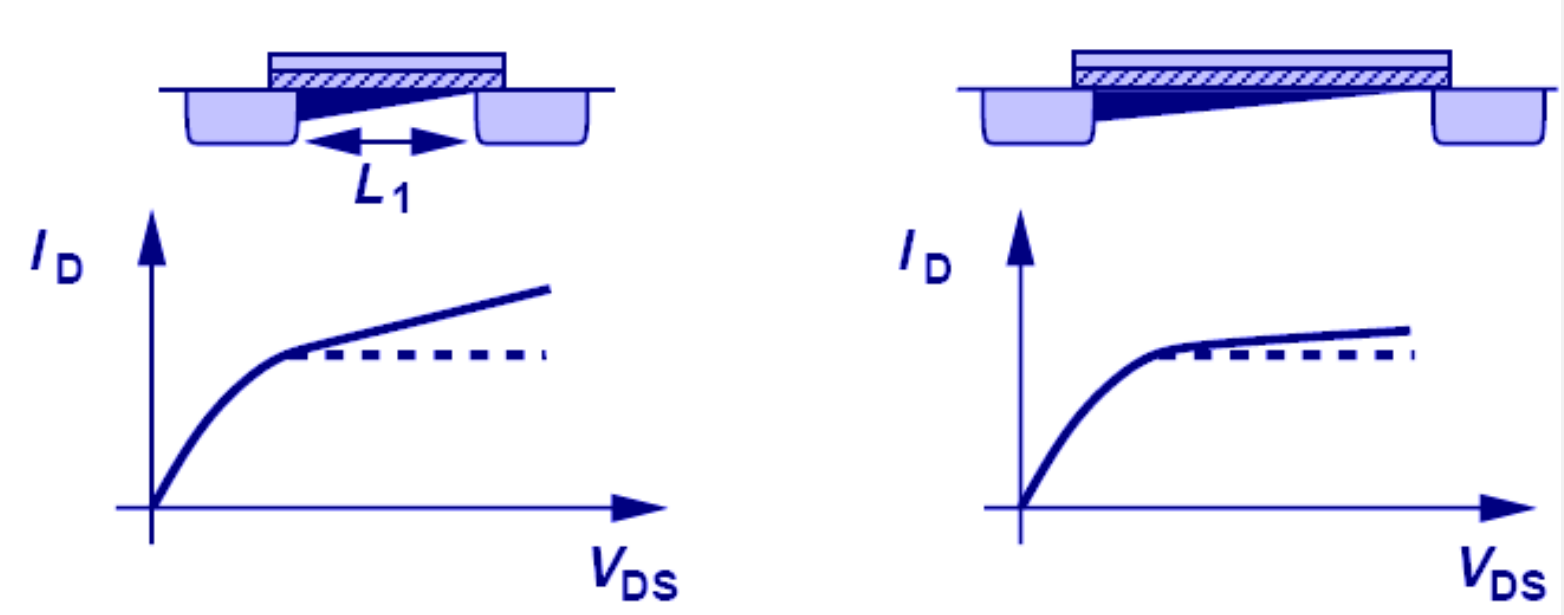


$$I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS}-V_{TH})^2(1+\lambda V_{DS})$$

λ: channel-length modulation coefficient

λ and L

- Unlike the Early voltage in BJT, the channel-length modulation factor can be controlled by the circuit designer.
- For long L, the channel-length modulation effect is less than that of short L.



$$I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L_1}(V_{GS} - V_{TH})^2$$

$$I_D \propto \frac{1}{L_1}$$

Transconductance

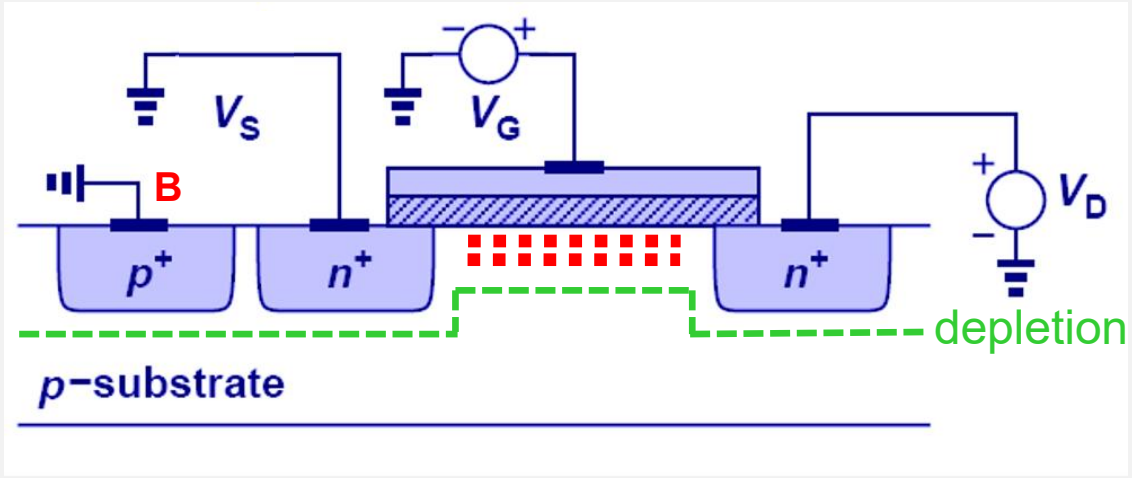
- Transconductance is a measure of how strong the drain current changes when the gate voltage changes.
- It has three different expressions.

$$(1)\ g_m = \mu_n C_{ox} \frac{W}{L}(V_{GS} - V_{TH}), \qquad (2)\ g_m = \sqrt{2\mu_n C_{ox} \frac{W}{L} I_D}\,, \qquad (3)\ g_m = \frac{2I_D}{V_{GS} - V_{TH}}$$

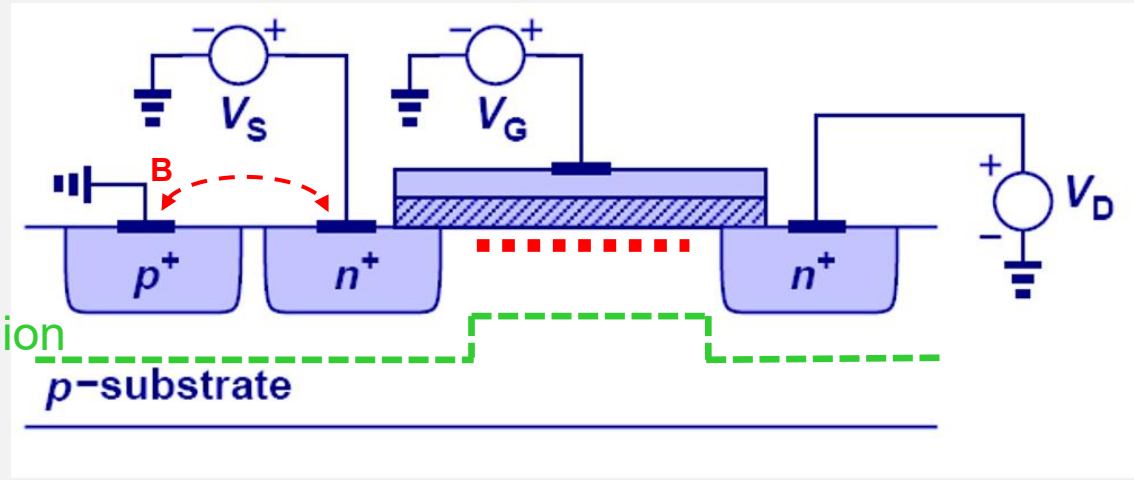
$\frac{W}{L}$ Constant $V_{GS} - V_{TH}$ Variable	$\frac{W}{L}$ Variable $V_{GS} - V_{TH}$ Constant	$\frac{W}{L}$ Variable I_D Constant
$g_m \propto \sqrt{I_D}$	$g_m \propto I_D$	$g_m \propto \sqrt{\frac{W}{L}}$
$g_m \propto V_{GS} - V_{TH}$	$g_m \propto \frac{W}{L}$	$g_m \propto \frac{1}{V_{GS} - V_{TH}}$

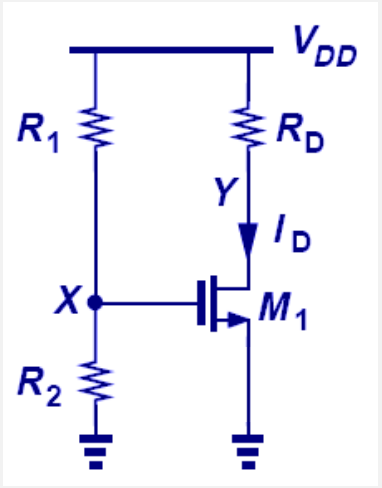
Body Effect

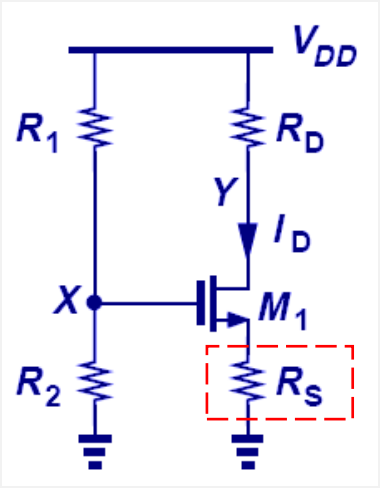
$V_{SB} = 0$



$V_{SB} > 0$

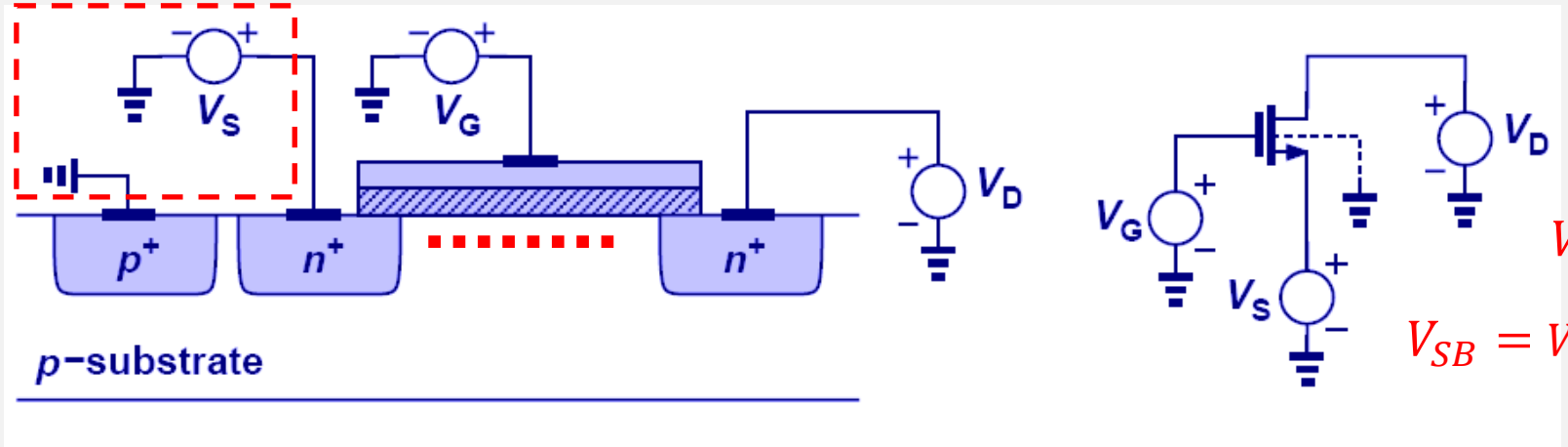






Body Effect

- As the source potential departs from the bulk potential, the threshold voltage changes.



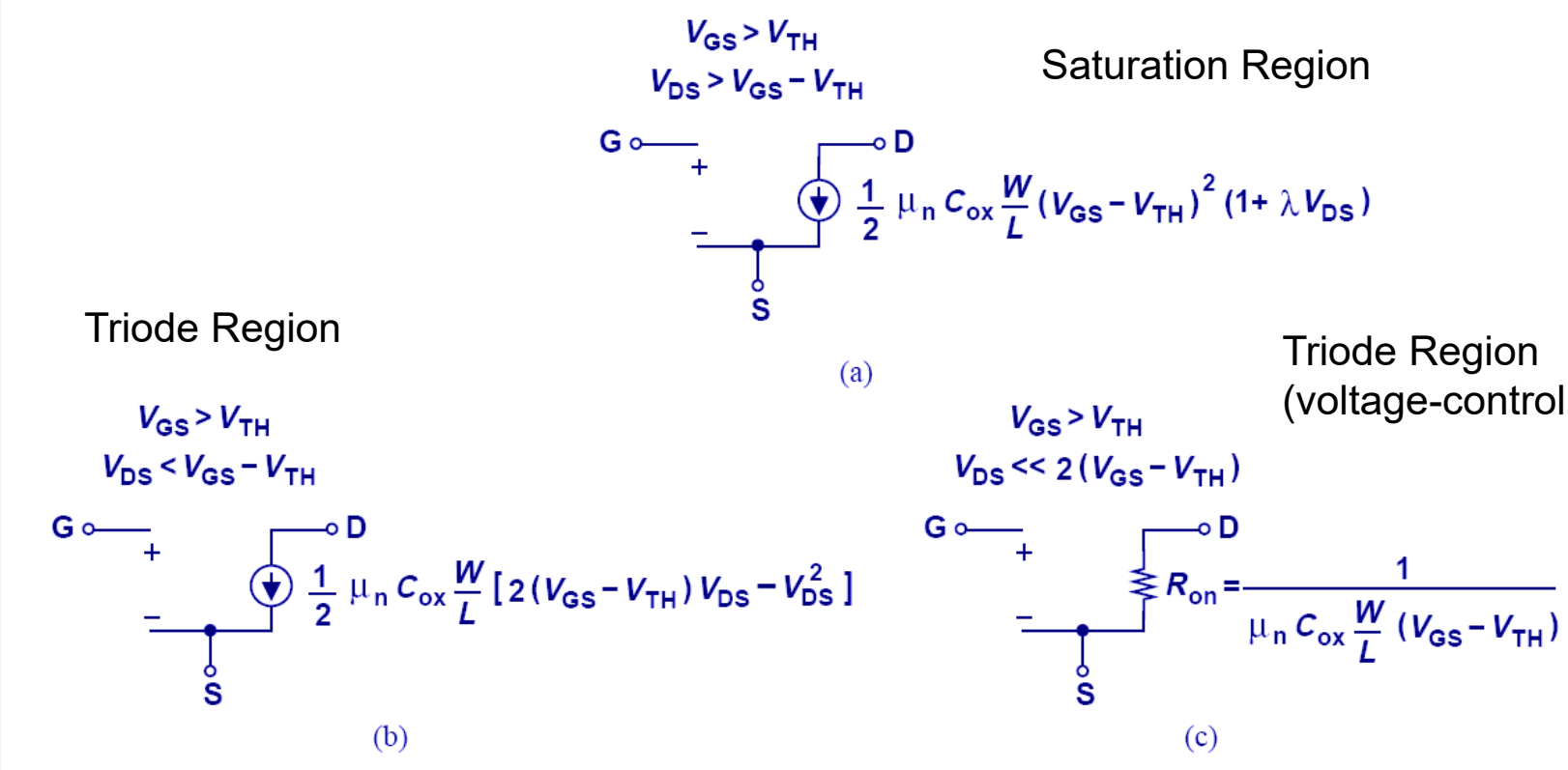
- In particular, as the source becomes more positive with respect to the substrate, V_{TH} increases. Called "**body effect**," this phenomenon is formulated as

$$V_{TH} = V_{TH0} + \rho\left(\sqrt{2\varphi_F + V_{SB}} - \sqrt{2\varphi_F}\right)$$

$$V_{SB} \uparrow \Rightarrow V_{TH} \uparrow$$

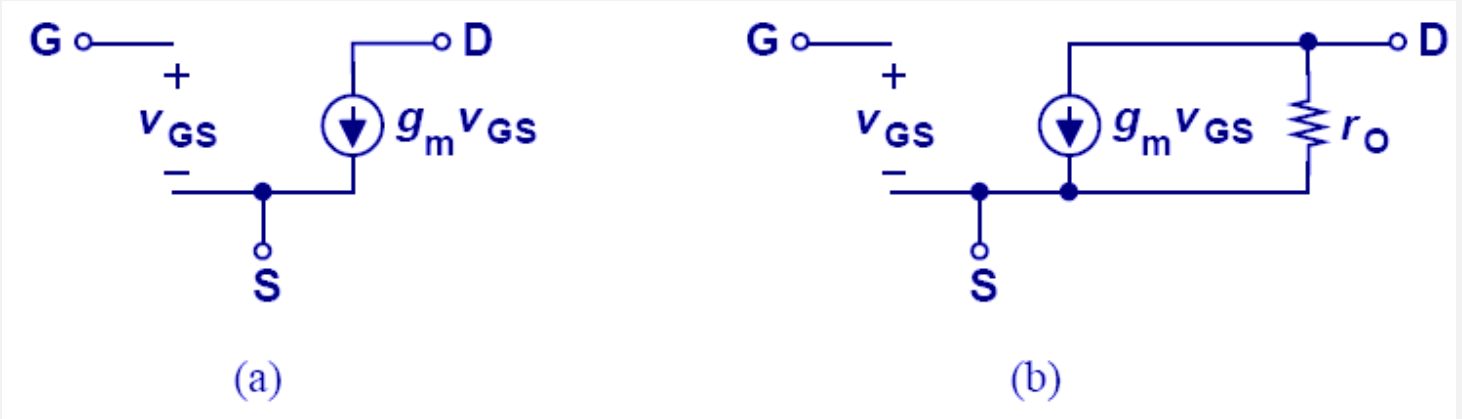
Large-Signal Models

- Based on the value of $V_{DS,}$ MOSFET can be represented with different large-signal models.



Small-Signal Model

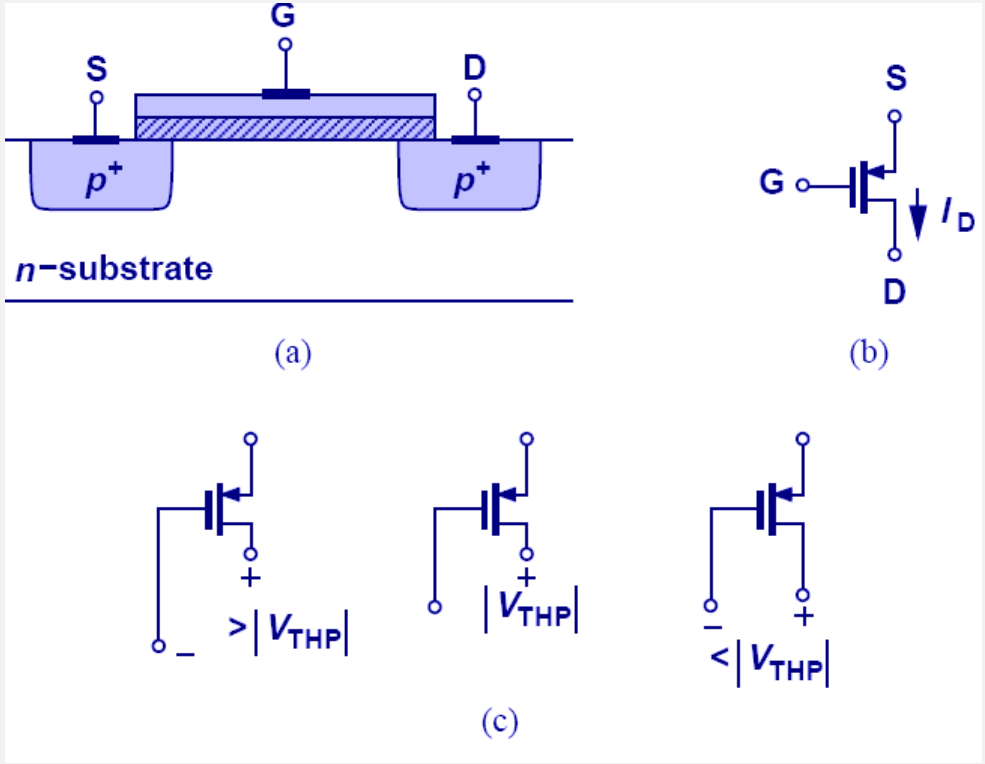
- When the bias point is not perturbed significantly, small-signal model can be used to facilitate calculations.
- To represent channel-length modulation, an output resistance is inserted into the model.



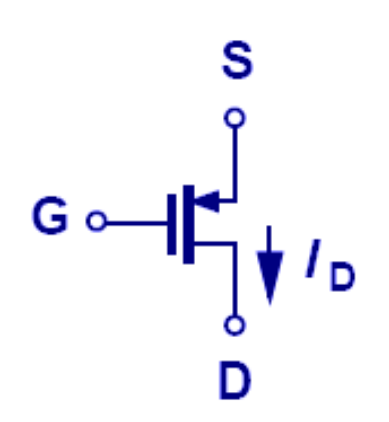
$$r_o \approx \frac{1}{\lambda I_D}$$

PMOS Transistor

- Just like the PNP transistor in bipolar technology, it is possible to create a MOS device where holes are the dominant carriers. It is called the PMOS transistor.
- It behaves like an NMOS device with all the polarities reversed.



PMOS Equations



$$I_{D,sat} = \frac{1}{2}\mu_p C_{ox}\frac{W}{L}(V_{GS} - V_{TH})^2(1 - \lambda V_{DS})$$

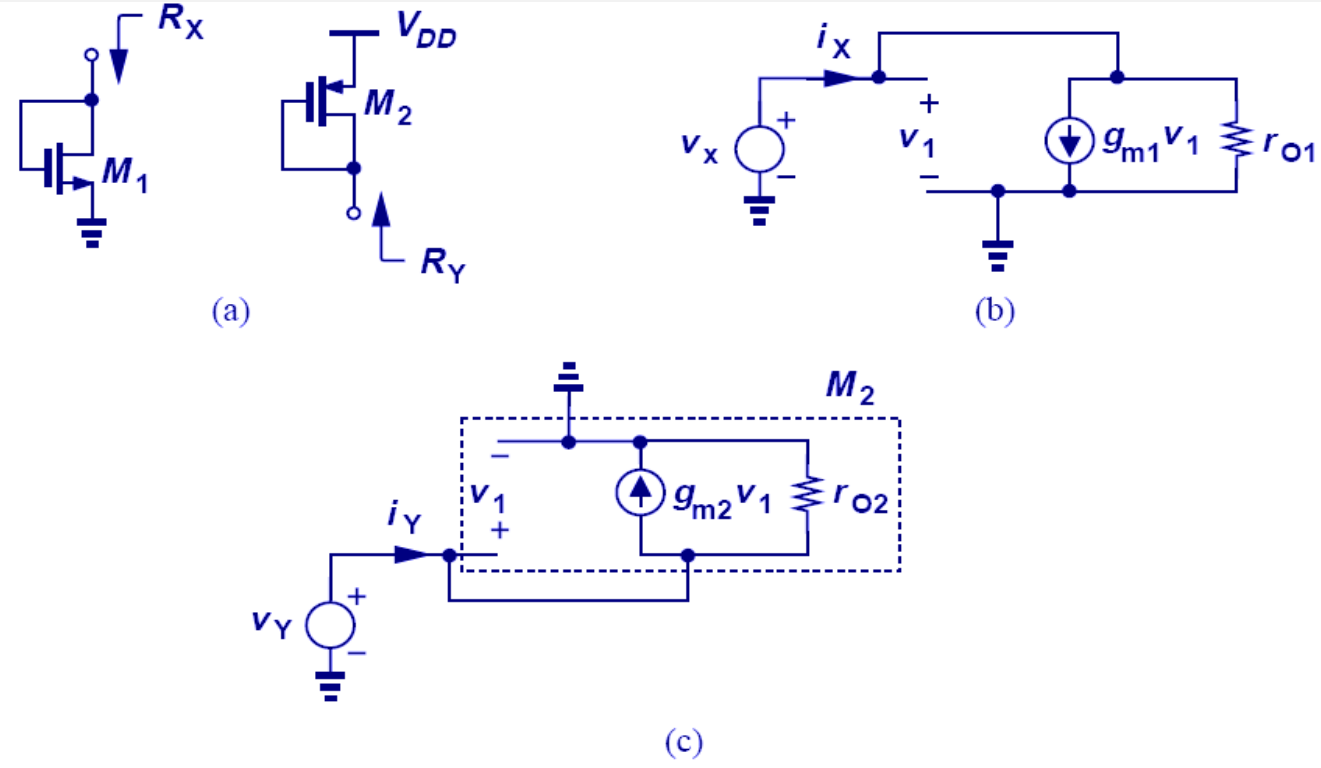
$$I_{D,tri} = \frac{1}{2}\mu_p C_{ox}\frac{W}{L}[2(V_{GS} - V_{TH})V_{DS} - V_{DS}^2]$$

$$I_{D,sat} = \frac{1}{2}\mu_p C_{ox}\frac{W}{L}(|V_{GS}| - |V_{TH}|)^2(1 + \lambda|V_{DS}|)$$

$$I_{D,tri} = \frac{1}{2}\mu_p C_{ox}\frac{W}{L}[2(|V_{GS}| - |V_{TH}|)|V_{DS}| - V_{DS}^2]$$

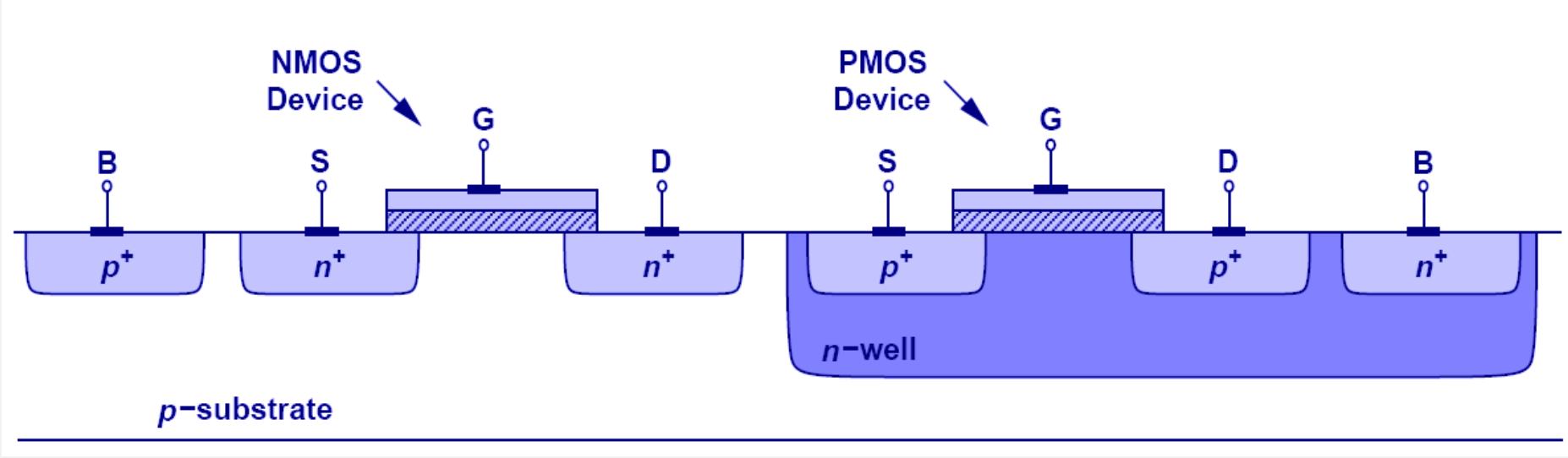
Small-Signal Model of PMOS Device

- The small-signal model of PMOS device is identical to that of NMOS transistor; therefore, R_X equals R_Y and hence $(1/gm)||r_o$.



CMOS Technology

- It possible to grow an n-well inside a p-substrate to create a technology where both NMOS and PMOS can coexist.
- It is known as CMOS, or “Complementary MOS”.



Comparison of Bipolar and MOS Transistors

- Bipolar devices have a higher g_m than MOSFETs for a given bias current due to its exponential IV characteristics.

Bipolar Transistor	MOSFET
Exponential Characteristic	Quadratic Characteristic
Active: $V_{CB} > 0$	Saturation: $V_{DS} > V_{GS} - V_{TH}$
Saturation: $V_{CB} < 0$	Triode: $V_{DS} < V_{GS} - V_{TH}$
Finite Base Current	Zero Gate Current
Early Effect	Channel−Length Modulation
Diffusion Current	Drift Current
–	Voltage−Dependent Resistor

info@em.lakewoodmillbarlett.com

$$I_C = I_S \exp \frac{V_{BE}}{V_T}$$

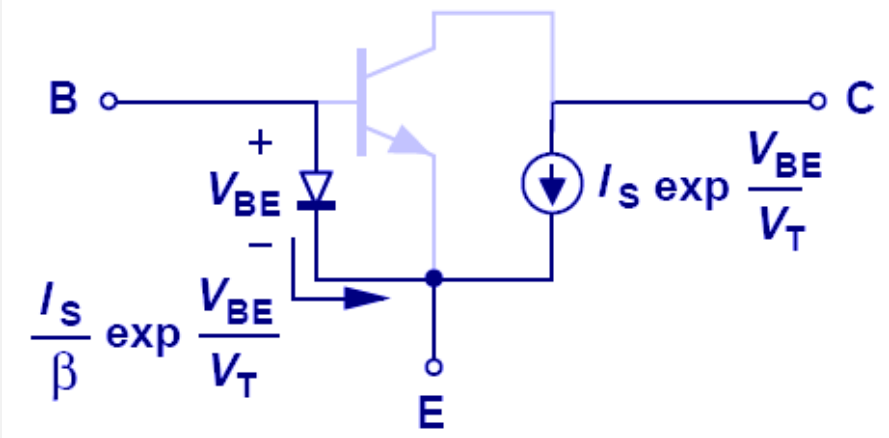
$$g_m = \frac{1}{V_T} I_S \exp \frac{V_{BE}}{V_T}$$

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})^2$$

Summary of Bipolar Transistor

Ch. 4 Bipolar Transistor Large-Signal Model

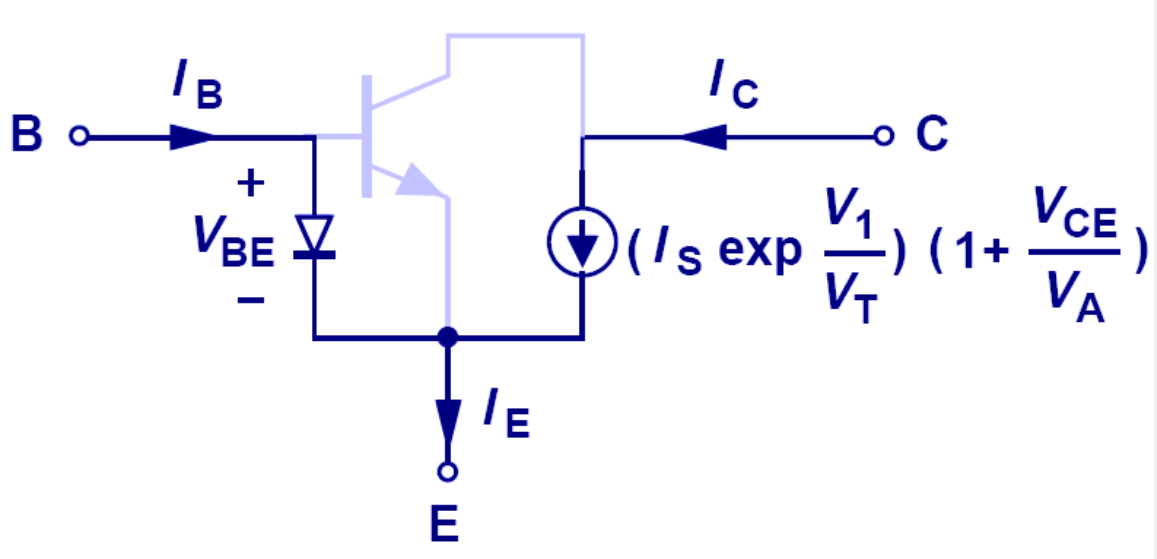
- A diode is placed between base and emitter and a voltage controlled current source is placed between the collector and emitter.
- $V_{BE} \longrightarrow I_C \longrightarrow I_B \longrightarrow I_E$: The sequence of current and voltage dependencies begins with the base-emitter voltage controlling the collector current, which is proportional to the base current, with their sum flowing through the emitter.



Large-signal model of bipolar transistor in active region.

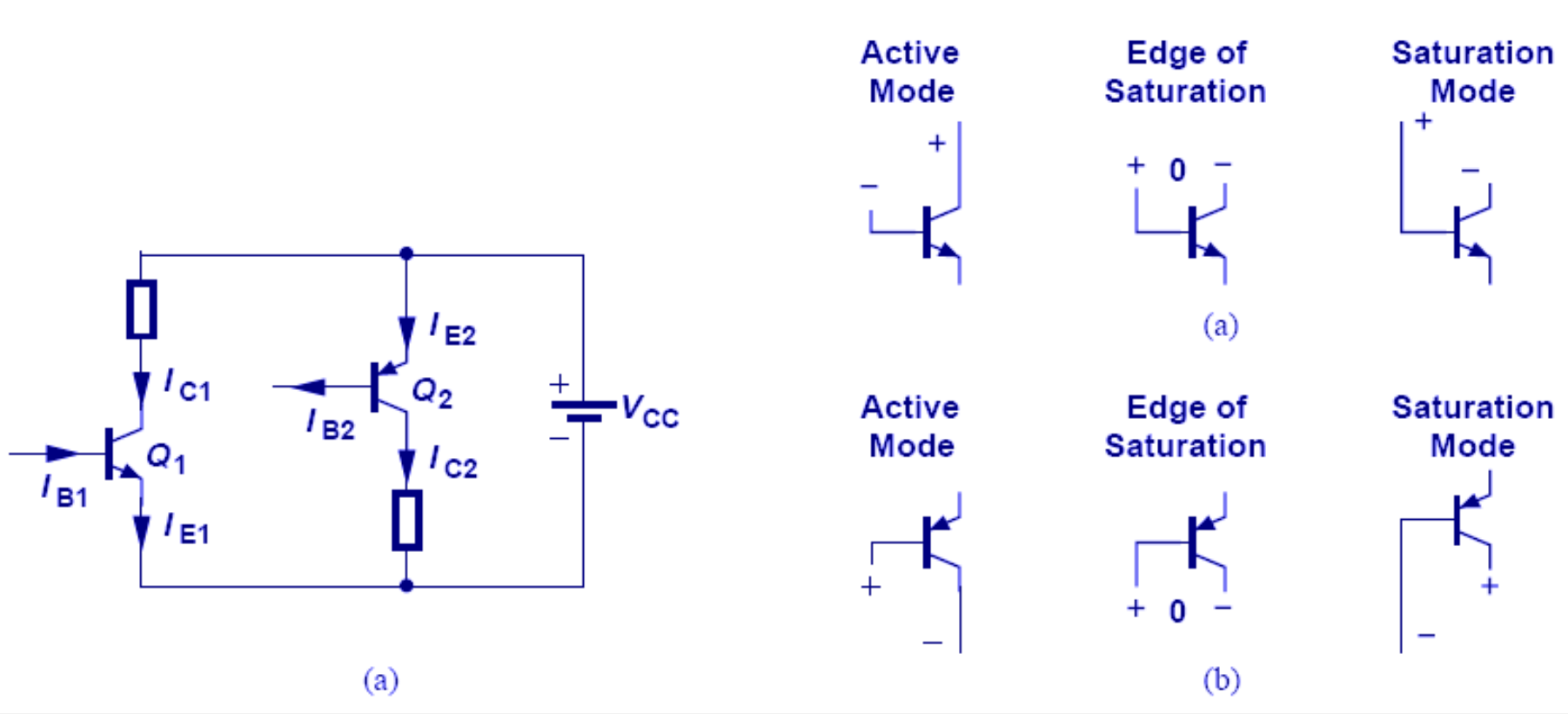
Ch. 4 Early Effect and Large-Signal Model

- Early effect can be accounted for in large-signal model by simply changing the collector current with a correction factor.
- In this mode, base current does not change.



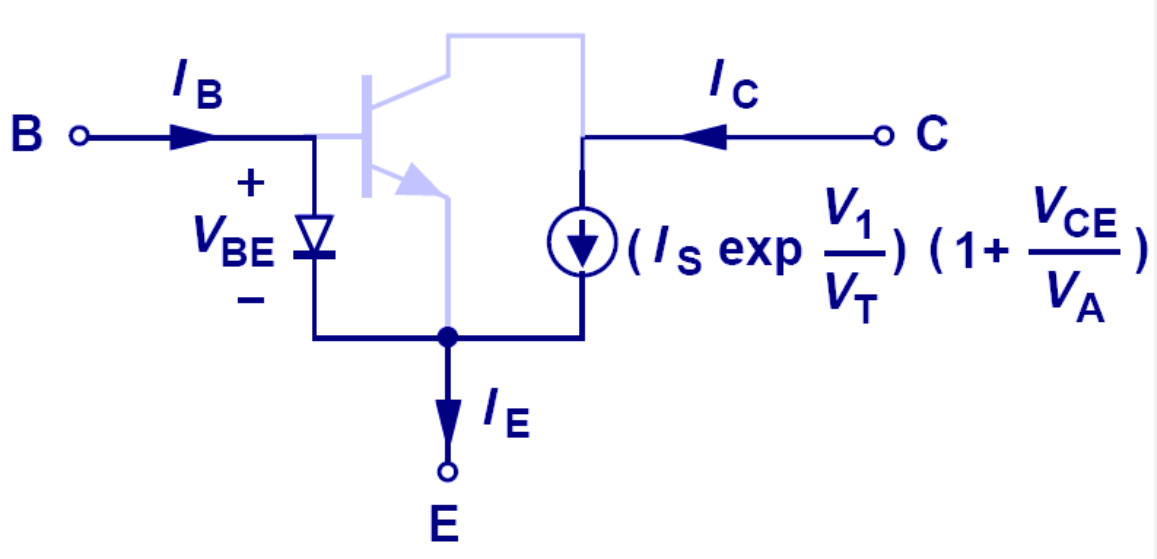
Ch. 4 A Comparison between NPN and PNP Transistors

- The figure below summarizes the direction of current flow and operation regions for both the NPN and PNP BJT's.

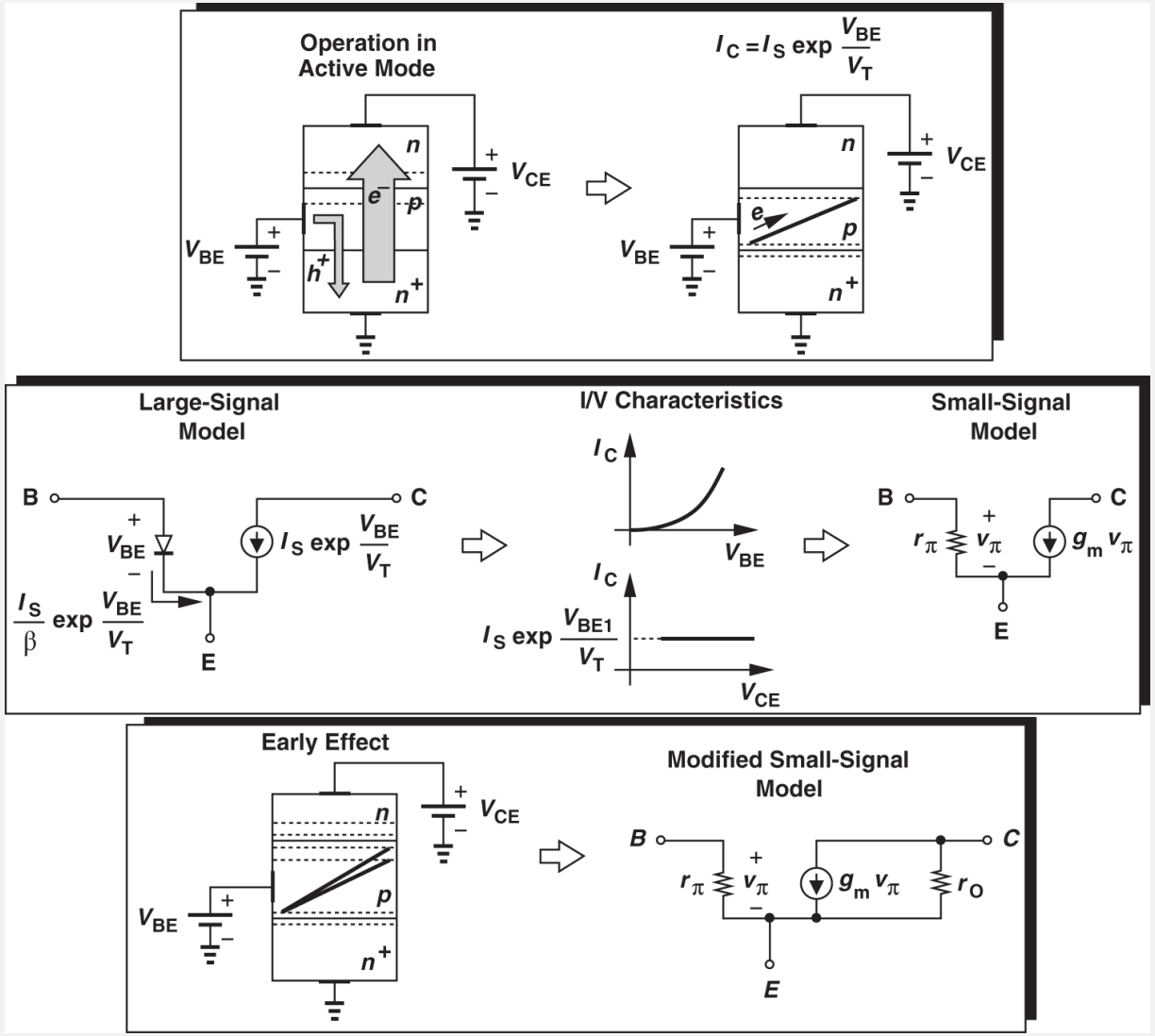


Ch. 4 Early Effect and Large-Signal Model

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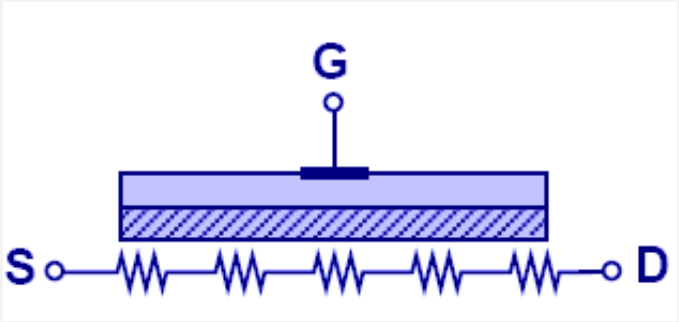
Ch. 4 Summary


Operation in Active Mode
$I_C = I_S \exp \frac{V_{BE}}{V_T}$
V_{CE}
V_{BE}
n
p
n^+
e^-
h^+
Large-Signal Model
I/V Characteristics
Small-Signal Model
B
C
E
$\frac{I_S}{\beta} \exp \frac{V_{BE}}{V_T}$
$I_S \exp \frac{V_{BE}}{V_T}$
I_C
V_{BE}
V_{CE}
$I_S \exp \frac{V_{BE1}}{V_T}$
r_π
v_π
$g_m v_\pi$
Early Effect
Modified Small-Signal Model
r_O

Supplement

Voltage-Dependent Resistor

- The inversion channel of a MOSFET can be seen as a resistor.
- Since the charge density inside the channel depends on the gate voltage, this resistance is also voltage-dependent.



Voltage-Controlled Attenuator

- As the gate voltage decreases, the output drops because the channel resistance increases.
- This type of gain control finds application in cell phones to avoid saturation near base stations.

